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Working with the Registry

We should be comfortable with the CLI at this point. It's time to level our skills again and tackle one of the more complicated aspects of the Windows operating system, the **Registry**. This section will walk us through what the Registry is, how to navigate it, and how to read key/value pairs and make changes to it as needed.

What Is The Windows Registry?

At its core, the **Registry** can be considered a hierarchal tree that contains two essential elements: **keys** and **values**. This tree stores all the required information for the operating system and the software installed to run under subtrees (think of them as branches of a tree). This information can be anything from settings to installation directories to specific options and values that determine how everything functions. As Pentesters, the Registry is a great spot to find helpful information, plant persistence, and more. [MITRE](#) provides many great examples of what a threat actor can do with access (locally or remotely) to a host's registry hive.

What are Keys

Keys, in essence, are containers that represent a specific component of the PC. Keys can contain other keys and values as data.

These entries can take many forms, and naming contexts only require that a Key be named using alphanumeric (printable) characters and is not case-sensitive. As a visual example of Keys, if we look at the image below, each folder within the **Green rectangle** is a Key and contains sub-keys.

Keys (Green)

The screenshot shows the Windows Registry Editor with the path **Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Adobe\Adobe Acrobat\10.0\Installer** selected. A green rectangle highlights the following hierarchy of keys:

- Computer
- HKEY_LOCAL_MACHINE
- SOFTWARE
- Adobe
- Adobe Acrobat
- 10.0
- Installer

The right pane shows the values for the selected key:

Name	Type	Data
(Default)	REG_SZ	(value not set)
DisableMaintenance	REG_DWORD	0x00000001 (1)



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Registry Key Files

A host systems Registry **root keys** are stored in several different files and can be accessed from `C:\Windows\System32\Config\`.

Along with these Key files, registry hives are held throughout the host in various other places.

Root Registry Keys

Working with the Registry

```
PS C:\htb> Get-ChildItem C:\Windows\System32\config\
```

Directory: C:\Windows\System32\config

Mode	LastWriteTime	Length	Name
d----	12/7/2019 4:14 AM		Journal
d----	12/7/2019 4:14 AM		RegBack
d----	4/28/2021 11:43 AM		systemprofile
d----	9/18/2021 12:22 AM		TxR
-a---	10/12/2022 10:06 AM	786432	BBI
-a---	1/20/2021 5:13 PM	28672	BCD-Template
-a---	10/18/2022 11:14 AM	38273024	COMPONENTS
-a---	10/12/2022 10:06 AM	1048576	DEFAULT
-a---	10/15/2022 9:33 PM	13463552	DRIVERS
-a---	1/27/2021 2:54 PM	32768	ELAM
-a---	10/12/2022 10:06 AM	131072	SAM
-a---	10/12/2022 10:06 AM	65536	SECURITY
-a---	10/12/2022 10:06 AM	168034304	SOFTWARE
-a---	10/12/2022 10:06 AM	29884416	SYSTEM
-a---	10/12/2022 10:06 AM	1623	VSMIDK

For a detailed list of all Registry Hives and their supporting files within the OS, we can look [HERE](#). Now let's discuss Values within the Registry.

What Are Values

Values represent data in the form of objects that pertain to that specific Key. These values consist of a name, a type specification, and the required data to identify what it's for. The image below visually represents **Values** as the data between the **Orange** lines. Those values are nested within the Installer key, which is, in turn, inside another key.

Values

Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Adobe\Adobe Acrobat\10.0\Installer

Name	Type	Data
(Default)	REG_SZ	(value not set)
DisableMaintenance	REG_DWORD	0x00000001 (1)



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We can reference the complete list of Registry Key Values [HERE](#). In all, there are 11 different value types that can be configured.

Registry Hives

Each Windows host has a set of predefined Registry keys that maintain the host and settings required for use. Below is a breakdown of each hive and what can be found referenced within.

Hive Breakdown

Name	Abbreviation	Description
HKEY_LOCAL_MACHINE	HKLM	This subtree contains information about the computer's physical state , such as hardware and operating system data, bus types, memory, device drivers, and more.
HKEY_CURRENT_CONFIG	HKCC	This section contains records for the host's current hardware profile . (shows the variance between current and default setups) Think of this as a redirection of the HKLM CurrentControlSet profile key.
HKEY_CLASSES_ROOT	HKCR	Filetype information, UI extensions, and backward compatibility settings are defined here.
HKEY_CURRENT_USER	HKCU	Value entries here define the specific OS and software settings for each specific user. Roaming profile settings, including user preferences, are stored under HKCU.
HKEY_USERS	HKU	The default User profile and current user configuration settings for the local computer are defined under HKU.

There are other predefined keys for the Registry, but they are specific to certain versions and regional settings in Windows. For more information on those entries and Registry keys in general, check out the documentation provided by [Microsoft](#)

Why Is The Information Stored Within The Registry Important?

As a pentester, the Registry can be a treasure trove of information that can help us further our engagements. Everything from what software is installed, current OS revision, pertinent security settings, control of Defender, and more can be found in the Registry. Can we find all of this information in other places? Yes. But there is no better single point to find all of it and have the ability to make widespread changes to the host simultaneously. From an offensive perspective, the Registry is hard for Defenders to protect. The hives are enormous and filled with hundreds of entries. Finding a singular change or addition among the hives is like hunting for a needle in a haystack (unless they keep solid backups of their configurations and host states). Having a general understanding of the Registry and where key values are within can help us take action quicker and for defenders spot any issues sooner.

How Do We Access the Information?

From the CLI, we have several options to access the Registry and manage our keys. The first is using [reg.exe](#). [Reg](#) is a dos executable explicitly made for use in managing Registry settings. The second is using the [Get-Item](#) and [Get-ItemProperty](#) cmdlets to read keys and values. If we wish to make a change, the use of [New-ItemProperty](#) will do the trick.

Querying Registry Entries



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Get-Item

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```
PS C:\htb> Get-Item -Path Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run | Select-Object Name, Value

Name                           Value
----                           -
SecurityHealth                 SecurityHealth
RtkAudUService                 RtkAudUService
WavesSvc                      WavesSvc
DisplayLinkTrayApp            DisplayLinkTrayApp
LogiOptions                    LogiOptions
Acrobat Assistant 8.0          Acrobat Assistant 8.0
(default)                      (default)
Focusrite Notifier             Focusrite Notifier
AdobeGCInvoker-1.0            AdobeGCInvoker-1.0
```

It's a simple output and only shows us the name of the services/applications currently running. If we wished to see each key and object within a hive, we could also use `Get-ChildItem` with the `-Recurse` parameter like so:

Recursive Search

Working with the Registry

```
PS C:\htb> Get-ChildItem -Path HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion -Recurse

Hive: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths
<SNIP>
Name                           Property
----                           -
7zFM.exe                       (default) : C:\Program Files\7-Zip\7zFM.exe
                                Path       : C:\Program Files\7-Zip\
Acrobat.exe                     (default) : C:\Program Files\Adobe\Acrobat DC\Acrobat\Acrobat.exe
                                Path       : C:\Program Files\Adobe\Acrobat DC\Acrobat\
AcrobatInfo.exe                (default) : C:\Program Files\Adobe\Acrobat DC\Acrobat\AcrobatInfo.exe
                                Path       : C:\Program Files\Adobe\Acrobat DC\Acrobat\
AcroDist.exe                   (default) : C:\Program Files\Adobe\Acrobat DC\Acrobat\Acrodist.exe
                                Path       : C:\Program Files\Adobe\Acrobat DC\Acrobat\
Ahk2Exe.exe                    (default) : C:\Program Files\AutoHotkey\Compiler\Ahk2Exe.exe
AutoHotkey.exe                 (default) : C:\Program Files\AutoHotkey\AutoHotkey.exe
chrome.exe                     (default) : C:\Program Files\Google\Chrome\Application\chrome.exe
                                Path       : C:\Program Files\Google\Chrome\Application
cmmgr32.exe                    CmNative   : 2
                                CmstpExtensionDll : C:\Windows\System32\cmcf32.dll
CNMNSST.exe                    (default) : C:\Program Files (x86)\Canon\IJ Network Scanner Selector EX\CNMNSST
                                Path       : C:\Program Files (x86)\Canon\IJ Network Scanner Selector EX
devenv.exe                     (default) : "C:\Program Files\Microsoft Visual
                                Studio\2022\Community\common7\ide\devenv.exe"
dfshim.dll                     UseURL    : 1
excel.exe                      (default) : C:\Program Files\Microsoft Office\Root\Office16\EXCEL.EXE
                                Path       : C:\Program Files\Microsoft Office\Root\Office16\
                                UseURL     : 1
                                SaveURL    : 1
fsquirt.exe                    DropTarget : {047ea9a0-93bb-415f-a1c3-d7aeb3dd5087}
IEDIAG.EXE                     (default) : C:\Program Files\Internet Explorer\IEDIAGCMD.EXE
                                Path       : C:\Program Files\Internet Explorer;
IEDIAGCMD.EXE                  (default) : C:\Program Files\Internet Explorer\IEDIAGCMD.EXE
                                Path       : C:\Program Files\Internet Explorer;
IEXPLORE.EXE                   (default) : C:\Program Files\Internet Explorer\IEXPLORE.EXE
                                Path       : C:\Program Files\Internet Explorer;
install.exe                    BlockOnTSNonInstallMode : 1
javaws.exe                     (default) : C:\Program Files\Java\jre1.8.0_341\bin\javaws.exe
                                Path       : C:\Program Files\Java\jre1.8.0_341\bin
licensemanagershell.exe        (default) : C:\Windows\System32\licensemanagershell.exe
```



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```
MSACCESS.EXE (default) : C:\Program Files\Microsoft Office\Root\Office16\MSACCESS.EXE
Path         : C:\Program Files\Microsoft Office\Root\Office16\
UseURL       : 1
msedge.exe   (default) : C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
Path         : C:\Program Files (x86)\Microsoft\Edge\Application

Hive: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths\msedge.exe

Name          Property
----          -
SupportedProtocols http  :
                https :
<SNIP>
```

Now we snipped the output because it is expanding and showing each key and associated values within the **HKLM:**
\SOFTWARE\Microsoft\Windows\CurrentVersion key. We can make our output easier to read using the **Get-ItemProperty** cmdlet.
Let's try that same query but with **Get-ItemProperty**.

Get-ItemProperty

Working with the Registry

```
PS C:\htb> Get-ItemProperty -Path Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run

SecurityHealth      : C:\Windows\system32\SecurityHealthSystray.exe
RtkAudUService      : "C:\Windows\System32\DriverStore\FileRepository\realtekservice.inf_amd64_85cff53207359d\RtkAudUService64.exe" -background
WavesSvc            : "C:\Windows\System32\DriverStore\FileRepository\wavesapo9de.inf_amd64_d350b8504310bbf5\avesSvc64.exe" -Jack
DisplayLinkTrayApp  : "C:\Program Files\DisplayLink Core Software\DisplayLinkTrayApp.exe" -basicMode
LogiOptions          : C:\Program Files\Logitech\LogiOptions\LogiOptions.exe /novi
Acrobat Assistant 8.0 (default) : "C:\Program Files\Adobe\Acrobat DC\Acrobat\Acrotray.exe"
Focusrite Notifier   : "C:\Program Files\Focusriteusb\Focusrite Notifier.exe"
AdobeGCInvoker-1.0   : "C:\Program Files (x86)\Common Files\Adobe\AdobeGCCClient\AGCInvokerUtility.exe"
PSPPath              : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
PSParentPath         : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion
PSChildName          : Run
PSPProvider           : Microsoft.PowerShell.Core\Registry
```

Now let's break this down. We issued the **Get-ItemProperty** command, specified our **path** as looking into the Registry, and specified the key **HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run**. The output provides us with the **name** of the services started and the **value** that was used to run them (the path they were executed from). This Registry key is used to **start** services/applications when a user **logs in** to the host. It is a great key to have visibility over and to keep in mind as a penetration tester. There are several versions of this key which we will discuss a little later in this section. Using **Get-ItemProperty** is much more readable than **Get-Item** was. When it comes to querying information, we can also use **Reg.exe**. Let's take a look at the output from that. We are going to look at a more straightforward key for this example.

Reg.exe

Working with the Registry

```
PS C:\htb> reg query HKEY_LOCAL_MACHINE\SOFTWARE\7-Zip
```



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We queried the `HKEY_LOCAL_MACHINE\SOFTWARE\7-Zip` key with `Reg.exe`, which provided us with the associated values. We can see that **two** values are set, `Path` and `Path64`, the `ValueType` is a `Reg_SZ` value which specifies that it contains a Unicode or ASCII string, and that value is the path to 7-Zip `C:\Program Files\7-Zip\`.

Finding Info In The Registry

For us as pentesters and administrators, finding data within the Registry is a must-have skill. This is where `Reg.exe` really shines for us. We can use it to search for keywords and strings like `Password` and `Username` through key and value names or the data contained. Before we put it to use, let's break down the use of `Reg Query`. We will look at the command string `REG QUERY HKCU /F "password" /t REG_SZ /S /K`.

- **Reg query**: We are calling on `Reg.exe` and specifying that we want to query data.
- **HKCU**: This portion is setting the path to search. In this instance, we are looking in all of `HKey_Current_User`.
- **/f "password"**: `/f` sets the pattern we are searching for. In this instance, we are looking for `"Password"`.
- **/t REG_SZ**: `/t` is setting the value type to search. If we do not specify, `reg query` will search through every type.
- **/s**: `/s` says to search through all subkeys and values recursively.
- **/k**: `/k` narrows it down to only searching through Key names.

Searching With Reg Query

Working with the Registry

```
PS C:\htb> REG QUERY HKCU /F "Password" /t REG_SZ /S /K

HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\Winlogon\PasswordExpiryNotification
    NotShownErrorTime      REG_SZ      08::23::24, 2022/10/19
    NotShownErrorReason    REG_SZ      GetPwdResetInfoFailed

End of search: 2 match(es) found.
```

Our results from this query could be more exciting, but it's still worth taking a look and using a similar search for other keywords and phrases like `Username`, `Credentials`, and `Keys`. We could be surprised by what we find. As we can see, querying registry keys and values is relatively easy. What if we want to set a new value or create a new key?

Creating and Modifying Registry Keys and Values

When dealing with the modification or creation of **new keys and values**, we can use standard PowerShell cmdlets like `New-Item`, `Set-Item`, `New-ItemProperty`, and `Set-ItemProperty` or utilize `Reg.exe` again to make the changes we need. Let's try and create a new Registry Key below. For our example, we will create a new test key in the `RunOnce` hive `HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce` named `TestKey`. By placing the key and value in `RunOnce`, after it executes, it will be deleted.

Scenario: We have landed on a host and can add a new registry key for persistence. We need to set a key named `TestKey` and a value of `C:\Users\htb-student\Downloads\payload.exe` that tells `RunOnce` to run our payload we leave on the host the next time the user logs in. This will ensure that if the host restarts or we lose access, the next time the user logs in, we will get a



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Working with the Registry

```
PS C:\htb> New-Item -Path HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\ -Name TestKey
```

Hive: HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce

Name	Property
----	-----
TestKey	

We now have a new key within the RunOnce key. By specifying the `-Path` parameter, we avoid changing our location in the shell to where we want to add a key in the Registry, letting us work from anywhere as long as we specify the absolute path. Let's set a Property and a value now.

Set New Registry Item Property

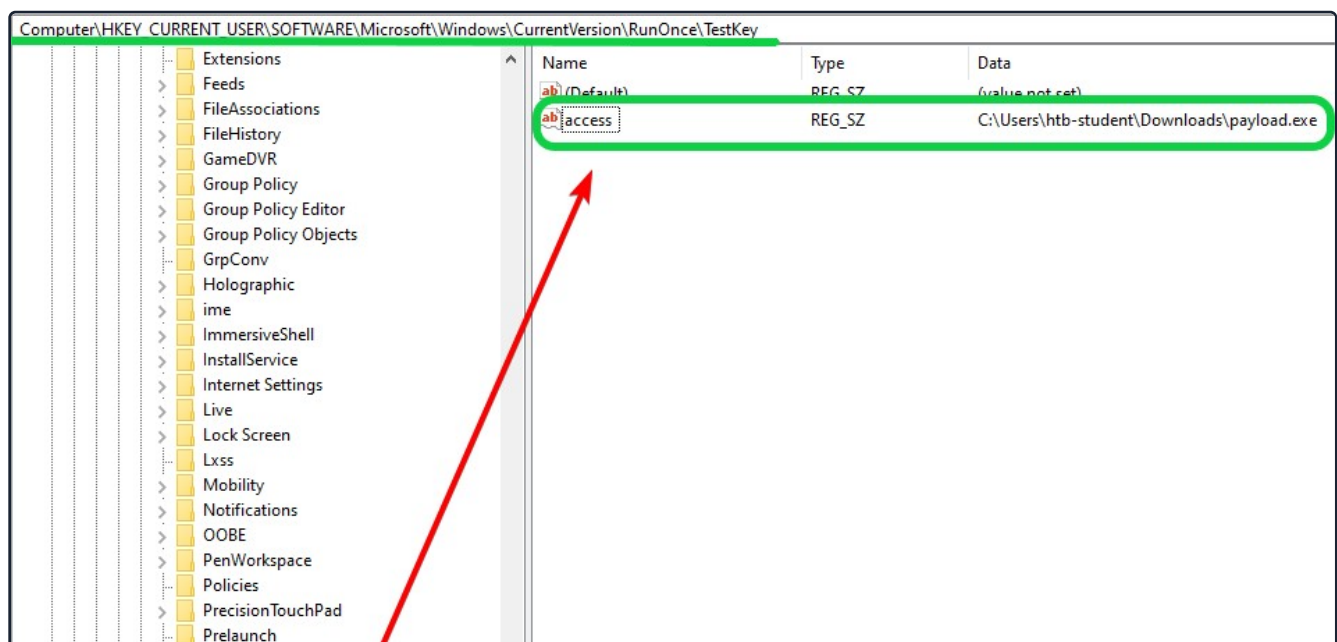
Working with the Registry

```
PS C:\htb> New-ItemProperty -Path HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\TestKey -Name "access"
```

```
access      : C:\Users\htb-student\Downloads\payload.exe
PSPath      : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\TestKey
PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce
PSChildName  : TestKey
PSDrive     : HKCU
PSProvider   : Microsoft.PowerShell.Core\Registry
```

After using `New-ItemProperty` to set our value named `access` and specifying the value as `C:\Users\htb-student\Downloads\payload.exe` we can see in the results that our value was created successfully, and the corresponding information, such as path location and Key name. Just to show that our key was created, we can see the new key and its values in the image below from the GUI Registry editor.

TestKey Creation





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If we wanted to add the same key/value pair using Reg.exe, we would do so like this:

Code: powershell

```
reg add "HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunOnce\TestKey" /v access /t REG_SZ /d "a"
```

Now in a real pentest, we would have left an executable payload on the host, and in the instance that the host reboots or the user logs in, we would acquire a new shell to our C2. This value doesn't do much for us right now, so let's practice deleting it.

Delete Reg properties

Working with the Registry

```
PS C:\htb> Remove-ItemProperty -Path HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\TestKey -Name "a"
PS C:\htb> Get-ItemProperty -Path HKCU:\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\TestKey
```

If no error window popped up, our key/value pair was deleted successfully. However, this is one of those things you should be extremely careful with. Removing entries from the Windows Registry could negatively affect the host and how it functions. Be sure you know what it is you are removing before. In the wise words of Uncle Ben, "With great power comes great responsibility."

Onwards

Now that we have Registry management down, it's time to move on to handling Event Logs through PowerShell.

VPN Servers

Warning: Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.

All VM instances associated with the old VPN Server will be terminated when switching to a new VPN server.

Existing PwnBox instances will automatically switch to the new VPN server.

PROTOCOL

☒ UDP 1337 ☐ TCP 443

DOWNLOAD VPN CONNECTION FILE



Connect to Pwnbox

Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU

10000ms ▼

Terminate Pwnbox to switch location



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Start Instance

∞ / 1 spawns left

Waiting to start...



Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): [Click here to spawn the target system!](#)



Cheat Sheet



Download VPN
Connection File

+0

A registry entry is made up of two pieces, a 'Key' and ' '. What is the second piece?

Submit your answer here...

+10 Streak pts



Submit



Hint



Show Solution

+0

What is the abbreviation for " HKey_Current_User".

Submit your answer here...

+10 Streak pts



Submit



Hint



Show Solution

+0

Take some time to practice adding and modifying the registry. Use the target host as a testbed and type "COMPLETE" as the answer below when you are done.

Submit your answer here...

+10 Streak pts



Submit



Hint



Show Solution



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My Workstation



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 Start Instance

 / 1 spawns left